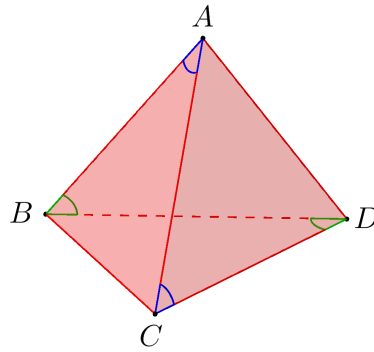


3D Geometry - Solutions

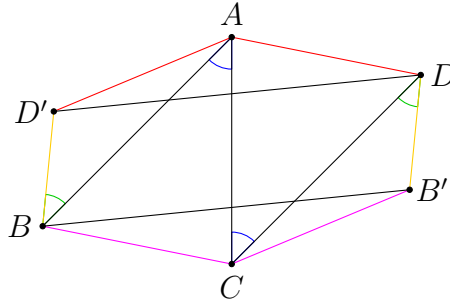
Problem 1. Given a convex polyhedron, the angular defect at a vertex is defined as 360° minus the sum of angles at this vertex. Prove that the total angular defect of a convex polyhedron is 720° .

Solution. See Theorem 1.3 of Three-dimensional Geometry 1.0.

Problem 2. Let $ABCD$ be a tetrahedron such that $\angle BAC = \angle ACD$ and $\angle ABD = \angle BDC$. Prove that $AB = CD$.



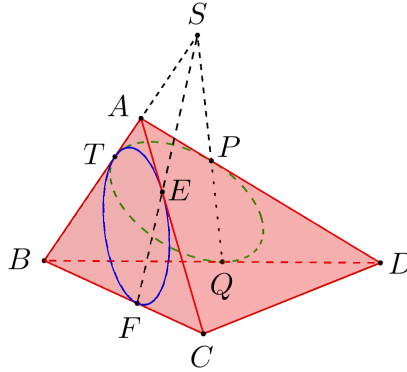
Solution. We cut the tetrahedron along the edges AD, DB, BC and unfold it to a two-dimensional figure. In this figure, $\triangle ABD'$, $\triangle ABC$, $\triangle ACD$ and $\triangle B'CD$ are the faces of the tetrahedron, so $AD' = AD$, $BD' = B'D$, $BC = B'C$, $\angle BAC = \angle DCA$ and $\angle ABD' = \angle CDB'$.



Since $\angle BAC = \angle DCA$, we have $AB \parallel DC$. Since $\angle ABD' = \angle CDB'$, we have $BD' \parallel B'D$. Together with $BD' = B'D$, we know that $BB'DD'$ is a parallelogram. Note that A and C lie on the perpendicular bisector of DD' and BB' respectively.

Now, B does not lie on the perpendicular bisector of DD' since $BD > BD'$ as a result of the unfold of tetrahedron. Thus, the position of A on the perpendicular bisector of DD' is uniquely determined by the value of $\angle ABD'$. The same holds for the position of C . It follows that A and C are symmetric points with respect to the common midpoint of BD and $B'D'$, since in that case $\angle ABD'$ and $\angle CDB'$ must be the same by symmetry. This implies $ABCD$ is a parallelogram, and hence $AB = CD$. This means $AB = CD$ in the original tetrahedron.

Problem 3. (Poland TST 1995 Second Round Problem 5) The incircles of the faces ABC and ABD of a tetrahedron $ABCD$ are tangent to the edge AB in the same point. Prove that the points of tangency of these incircles to the edges AC, BC, AD, BD are concyclic.



Solution. Let T be the common tangent point of the incircles and AB . Let the incircle of $\triangle ABC$ touch AC and BC at E and F , and let the incircle of $\triangle ABD$ touch AD and BD at P and Q respectively. If T is the midpoint of AB , then $EF \parallel AB \parallel PQ$, and hence $EFQP$ forms an isosceles trapezoid by symmetry. So E, F, Q, P are concyclic.

If T is not the midpoint of AB , then EF meets AB at a point S such that $(A, B; S, T)$ is a harmonic division (this means $\frac{SA}{SB} = \frac{TA}{TB}$ with S lying outside the segment AB , which can be deduced by Menelaus' theorem). Similarly, PQ also meets AB at S . Thus, E, F, Q, P are coplanar. It follows by the power of a point theorem that

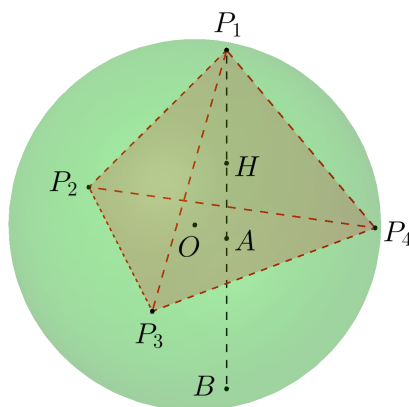
$$SE \times SF = ST^2 = SP \times SQ.$$

This implies E, F, Q, P are concyclic.

Problem 4. Let $ABCD$ be a tetrahedron. Suppose the medians from A of $\triangle ABC$, $\triangle ABD$ and $\triangle ACD$ are pairwise perpendicular. Prove that $AB = AC = AD$.

Solution. See Problem 3.1 of Three-dimensional Geometry 1.0.

Problem 5. (HMIC 2020 Problem 3) Let $P_1P_2P_3P_4$ be a tetrahedron in $\mathbb{R}^3$ and let O be a point equidistant from each of its vertices. Suppose there exists a point H such that for each i , the line P_iH is perpendicular to the plane through the other three vertices. Line P_1H intersects the plane through P_2, P_3, P_4 at A , and contains a point $B \neq P_1$ such that $OP_1 = OB$. Show that $HB = 3HA$.

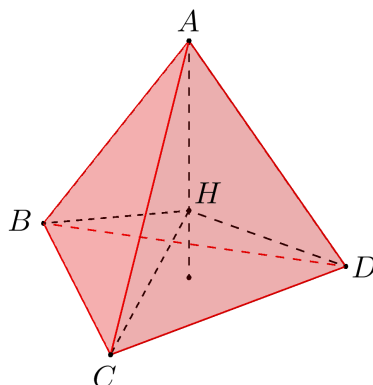


Solution. The conditions mean that $P_1P_2P_3P_4$ is an orthocentric tetrahedron, with H being the orthocentre. The point B lies on the circumsphere of $P_1P_2P_3P_4$. For an orthocentric tetrahedron, the foot of altitude A lies on the twelve-point sphere (it is a point Q_i in Three-dimensional Geometry 1.0). As the orthocentre H (which is the Monge point) is the centre of homothety sending the circumsphere to the twelve-point sphere with ratio $\frac{1}{3}$, the homothety maps B to A and hence $HB = 3HA$.

Problem 6. (IMO Longlist 1985) The sphere inscribed in tetrahedron $ABCD$ touches the sides ABD and DBC at points K and M , respectively. Prove that $\angle AKB = \angle DMC$.

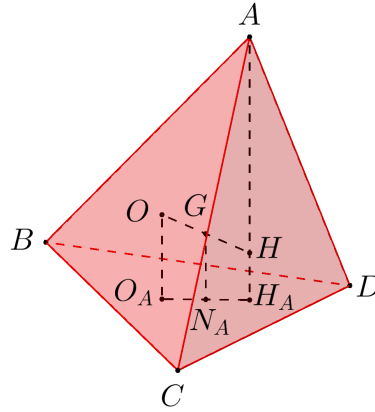
Solution. See Problem 2.5 of Three-dimensional Geometry 1.0.

Problem 7. Let $ABCD$ be an orthocentric tetrahedron with orthocentre H . Prove that $HBCD$ is orthocentric, and A is the orthocentre of $HBCD$.



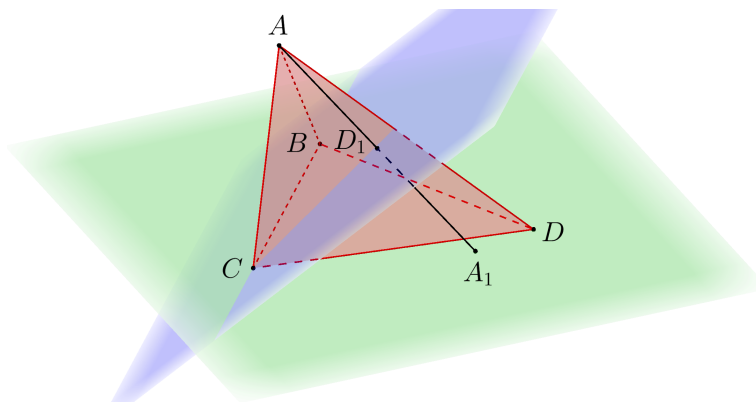
Solution. Firstly, AH is perpendicular to the plane BCD by definition. This shows A lies on the altitude from H of the tetrahedron $HBCD$. Secondly, AB is perpendicular to both HC and HD , which implies AB is perpendicular to the plane HCD . Therefore, A also lies on the altitude from B of the tetrahedron $HBCD$. By symmetry, A lies on the altitudes from C and D as well. So $HBCD$ is orthocentric, and A is the orthocentre of $HBCD$.

Problem 8. Let $ABCD$ be an orthocentric tetrahedron. Let ℓ_A be the line passing through the nine-point centre of $\triangle BCD$ and perpendicular to the plane BCD . Similarly, define ℓ_B, ℓ_C and ℓ_D . Prove that ℓ_A, ℓ_B, ℓ_C and ℓ_D are concurrent.



Solution. Let H, G, O be the orthocentre, centroid and circumcentre of $ABCD$. Let H_A, N_A, O_A be the projection of H, G, O on the plane BCD . Recall that H_A and O_A are the orthocentre and circumcentre of $\triangle BCD$. Since G is the midpoint of HO , N_A is the midpoint of H_AO_A , and hence N_A is the nine-point centre of $\triangle BCD$. It follows that G lies on ℓ_A . By symmetry, ℓ_A, ℓ_B, ℓ_C and ℓ_D are concurrent at G .

Problem 9. (Oral Moscow Team MO 2012 Grade 11 Problem 2) Perpendiculars AB_1, AC_1, AD_1 were dropped in the tetrahedron $ABCD$ from the vertex A on the plane dividing the dihedral angles at the edges CD, BD, BC in half. Prove that the plane $B_1C_1D_1$ is parallel to the plane BCD .



Solution. Let A_1 be the reflection of A in the internal bisecting plane with respect to BC . Then A_1 lies on the plane BCD . Thus, the midpoint D_1 of AA_1 lies on the plane Π which is equidistant to A and plane BCD . Similarly, B_1 and C_1 lie on Π , and thus the plane $B_1C_1D_1$ is just Π . Clearly, Π is parallel to BCD .

Problem 10. Let $ABCD$ be an orthocentric tetrahedron. Let H_B, H_C, H_D be the feet of altitudes from B, C, D respectively. Prove that B, C, D, H_B, H_C, H_D are cospherical.

Solution. See Problem 3.2 of Three-dimensional Geometry 1.0.

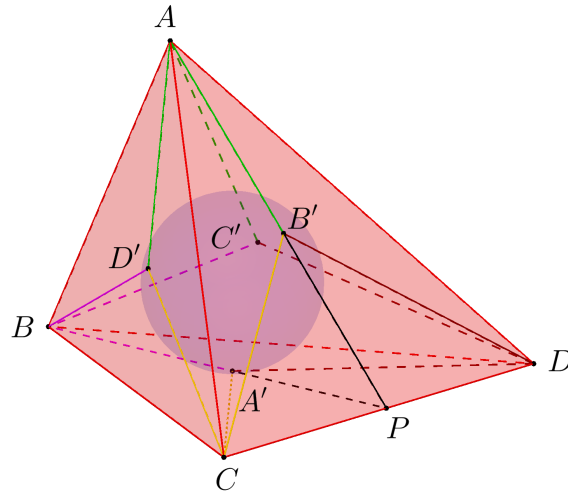
Problem 11. (All-Siberian Open School Olympiad 2013 Grade 11 Problem 4) Point M is marked on a sphere. Consider all triples of points A, B, C on the sphere other than M , such that the segments MA, MB, MC are pairwise perpendicular, and for each such triple we consider the plane passing through A, B, C . Prove that all such planes pass through a common point.

Solution. See Problem 2.3 of Three-dimensional Geometry 1.0.

Problem 12. Let R and r be the circumradius and inradius of a tetrahedron $ABCD$. Prove that $R \geq 3r$.

Solution. The twelve-point sphere of $ABCD$ has radius $\frac{R}{3}$. As it intersects all faces of $ABCD$, while the insphere is the smallest sphere inside $ABCD$ which intersects all faces of $ABCD$, we have $\frac{R}{3} \geq r$, i.e. $R \geq 3r$.

Problem 13. The insphere of a tetrahedron $ABCD$ touches its faces at points A', B', C', D' respectively. If the segments AA' and BB' intersect, prove that the segments CC' and DD' also intersect.



Solution. Since AA' and BB' intersect, A, B, A', B' are coplanar. As AB' and BA' cannot be parallel, they must intersect at a point P . Since CD is the intersection of the planes ACD and BCD , the point P must lie on CD . Thus,

$$\frac{CA' \sin \angle BA'C}{DA' \sin \angle BA'D} = \frac{CP}{DP} = \frac{CB' \sin \angle AB'C}{DB' \sin \angle AB'D}.$$

As $CA' = CB'$ and $DA' = DB'$, this yields

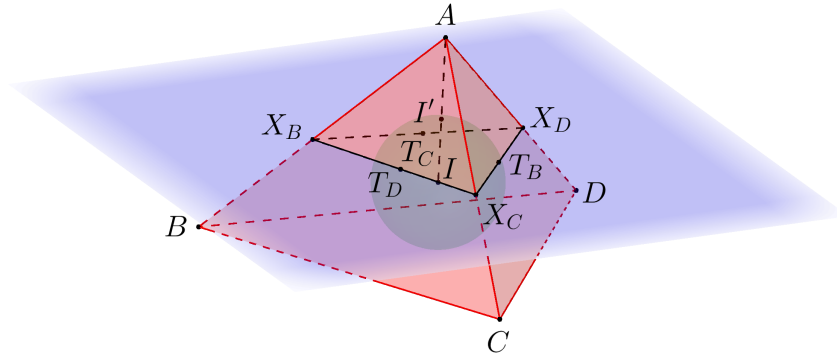
$$\frac{\sin \angle BA'C}{\sin \angle BA'D} = \frac{\sin \angle AB'C}{\sin \angle AB'D}. \quad (1)$$

Note that the steps are reversible. Therefore, by symmetry, CC' intersects DD' if and only if

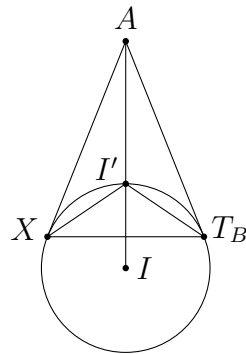
$$\frac{\sin \angle DC'A}{\sin \angle DC'B} = \frac{\sin \angle CD'A}{\sin \angle CD'B}. \quad (2)$$

Note that $\angle DC'A = \angle AB'D$, $\angle DC'B = \angle BA'D$, $\angle CD'A = \angle AB'C$ and $\angle CD'B = \angle BA'C$. Therefore, (2) follows from (1). This completes the proof.

Problem 14. Let I be the incentre of a tetrahedron $ABCD$. The insphere of $ABCD$ touches the faces ABC , ABD , ACD at T_D , T_C , T_B respectively. The plane $T_B T_C T_D$ cuts the edges AB , AC , AD at X_B , X_C , X_D respectively. Prove that the incentre of $AX_B X_C X_D$ is the intersection of the segment AI and the insphere of $ABCD$.



Solution. Since AI is the axis of the cone of tangents from A , it is perpendicular to the plane $T_B T_C T_D$. Consider the plane passing through A, I, T_B . It cuts the insphere in a great circle Γ . Let the tangent from A to Γ different from AT_B meet Γ at X . Then X lies on the plane $T_B T_C T_D$ since it is the touching point of a tangent from A to the insphere.



Now, it is clear that $\angle AT_B I' = \angle T_B X I' = \angle I' T_B X$. This implies $I' T_B$ is the internal angle bisector of $\angle AT_B X$. As $X_C X_D$ is perpendicular to the plane $A I T_B$ (it is perpendicular to both AI and IT_B), this shows I' lies on the internal bisecting plane of the dihedral angle between the planes $A X_C X_D$ and $X_B X_C X_D$. By symmetry, I' also lies on the internal bisecting planes with respect to $X_B X_C$ and $X_B X_D$. Therefore, I' is the incentre of $A X_B X_C X_D$.

Problem 15. Let I be the incentre of a tetrahedron $ABCD$. The insphere touches the plane BCD at T . Let I' be the antipode of I on the circumsphere of $IBCD$. The line $I'T$ meets the insphere of $ABCD$ again at P . Prove that the circumsphere of $PBCD$ is tangent to the insphere of $ABCD$.

Solution. See Problem 2.7 of Three-dimensional Geometry 1.0.

Problem 16. (Germany MO 2020 Problem 6) The insphere and the exsphere opposite to the vertex D of a tetrahedron $ABCD$ touch the face ABC in the points X and Y , respectively. Show that $\angle XAB = \angle CAY$.

Solution. See P.49 of Three-dimensional Geometry 1.0.

Problem 17. Let $ABCD$ be a tetrahedron. The points $P_{AB}, P_{AC}, P_{AD}, P_{BC}, P_{BD}$ and P_{CD} are chosen on the edges AB, AC, AD, BC, BD, CD respectively. Prove that the circumspheres of $AP_{AB}P_{AC}P_{AD}, BP_{AB}P_{BC}P_{BD}, CP_{AC}P_{BC}P_{CD}, DP_{AD}P_{BD}P_{CD}$ has a common point.

Solution. See Problem 2.4 of Three-dimensional Geometry 1.0.

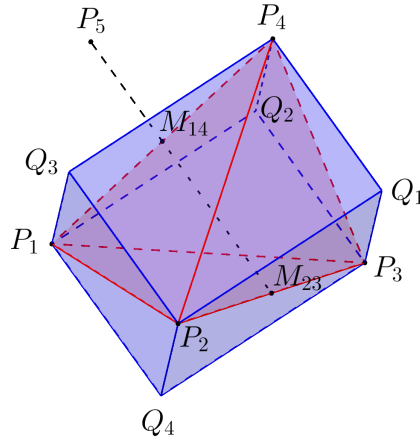
Problem 18. (SMMC 2020 A4) A regular spatial pentagon consists of five points P_1, P_2, P_3, P_4, P_5 in $\mathbb{R}^3$ such that $|P_i P_{i+1}| = |P_j P_{j+1}|$ and $\angle P_{i-1} P_i P_{i+1} = \angle P_{j-1} P_j P_{j+1}$ for all $1 \leq i, j \leq 5$, where $P_0 = P_5$ and $P_6 = P_1$. A regular spatial pentagon is planar if there is a plane passing through all five points P_1, P_2, P_3, P_4, P_5 .

Show that every regular spatial pentagon is planar.

Solution. Suppose on the contrary that P_1, P_2, P_3, P_4, P_5 are not coplanar. From the conditions, it is clear that $\triangle P_i P_{i+1} P_{i+2} \cong \triangle P_j P_{j+1} P_{j+2}$ for any i, j , where the indices are taken modulo 5. This implies $P_i P_{i+2} = P_j P_{j+2}$ for any i, j .

WLOG assume $P_1 P_2 P_3 P_4$ is non-degenerate. Consider the circumscribed parallelepiped of this tetrahedron. Let Q_1, Q_2, Q_3, Q_4 be the vertices opposite to P_1, P_2, P_3, P_4 respectively. Since it is given that $P_1 P_2 = P_3 P_4$, we have $P_1 P_2 = Q_3 Q_4$. Thus, the parallelogram $P_1 Q_4 P_2 Q_3$ is indeed a rectangle. Similarly, from $P_1 P_3 = P_2 P_4$, the parallelogram $P_2 Q_1 P_4 Q_3$ is also a rectangle.

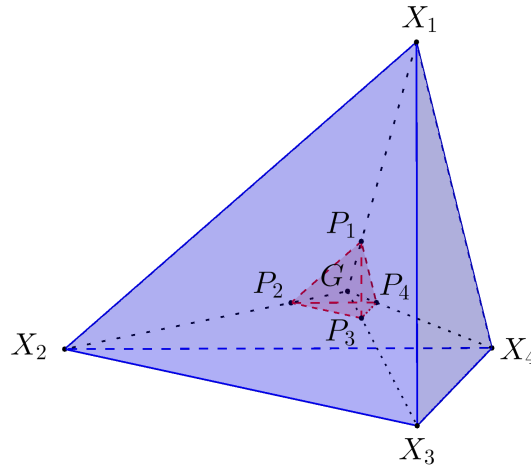
Let M_{23} and M_{14} be the midpoints of $P_2 P_3$ and $P_1 P_4$ respectively. As $M_{14} M_{23} // Q_3 P_2$, the segment $M_{14} M_{23}$ is perpendicular to both $P_2 P_3$ and $P_1 P_4$ from above. Therefore, $M_{23} M_{14}$ is the line of



intersection of the mediators of P_2P_3 and P_1P_4 . As $P_5P_1 = P_5P_4$ and $P_5P_2 = P_5P_3$, the point P_5 lies on both the mediators of P_2P_3 and P_1P_4 . Therefore, P_5 lies on $M_{23}M_{14}$.

Now, it is obvious that P_2, P_3, P_4, P_5 are not coplanar since the intersection of the line $M_{14}M_{23}$ and the plane $P_2P_3P_4$ is M_{23} . So we can repeat the same argument and prove inductively that each P_k lies on the line joining $M_{(k-1)(k+1)}$ and $M_{(k-2)(k+2)}$, where $M_{ij} = M_{ji}$ is the midpoint of P_i and P_j .

Consider the homothety using the centroid G of $P_1P_2P_3P_4$ as centre and with ratio 5. Let the image of $P_1P_2P_3P_4$ be $X_1X_2X_3X_4$. Since G cuts a median in the ratio 3 : 1 with G closer to the face, the distance from P_1 to the plane $X_2X_3X_4$ is twice the distance from P_1 to the plane $P_2P_3P_4$. Similar results hold for other vertices.



As P_4, M_{35}, M_{12} are collinear, M_{35} lies on the plane $P_1P_2P_4$, and so P_5 lies on the plane $X_1X_2X_4$. Similarly, P_5 lies on the plane $X_1X_3X_4$. As P_3, M_{24}, M_{15} are collinear, M_{15} lies on the plane $P_2P_3P_4$, and so P_5 lies on the plane $X_2X_3X_4$. Similarly, P_5 lies on the plane $X_1X_2X_3$. But $X_1X_2X_3X_4$ is non-degenerate since it is similar to $P_1P_2P_3P_4$. It is impossible to find a point lying on the planes of all its faces. This yields a contradiction.

Therefore, P_1, P_2, P_3, P_4, P_5 are coplanar.

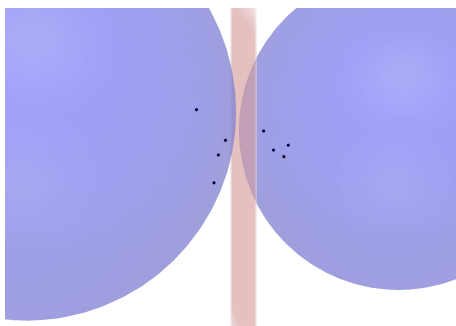
Combinatorial Geometry

Problem 19. There are 2020 different planes in the space. Prove that there exists a sphere intersecting exactly 100 of these planes.

Solution. The planes determine a finite number of bisecting planes of the dihedral angles between some pairs of planes. We choose a point O which does not lie on any of these bisecting planes. Consider a sphere with centre O . When we gradually increase the radius of this sphere, it intersects more and more planes. The number of planes that it intersects must be increased by 1 each time when the value increases, because the distances from O to different planes are different. Thus, there must be an instant when the sphere meets exactly 100 planes.

Problem 20. There are 2020 distinct points in $\mathbb{R}^3$. Prove that there are two non-intersecting spheres each containing exactly 1010 of these points.

Solution. Consider a plane Π which is not parallel to any line formed by 2 of the given points such that all the 2020 points lie on the same side of Π . We gradually move Π towards the points in a parallel direction. Each time Π only passes through at most one of the given points. Thus, there is an instant when Π has passed through exactly 1010 points. At this instant, each side of Π has exactly 1010 points.



Then we can find a sphere on each side of Π which contains all 1010 points on that side, so that the two spheres do not intersect, and each of them contains 1010 points. This is possible. For example, we choose the centre to be a point sufficiently far away from Π and choose the radius such that the sphere is almost tangent to Π . When the centre is further and further away, the sphere is ‘getting closer and closer’ to Π . Eventually, all points must be included in the sphere.

Problem 21. (Putnam 2002 A2) Given any five points on a sphere, show that some four of them must lie on a closed hemisphere.

Solution. Let A and B be two of the points. Consider a great circle containing A and B . By the pigeonhole principle, 2 of the remaining 3 points, say C and D , must be contained on the same closed hemisphere determined by the great circle. Then A, B, C, D lie on the same closed hemisphere.

Problem 22. (Putnam 2002 B2) Consider a convex polyhedron with at least five faces such that exactly three edges emerge from each of its vertices. Two players play the following game: Each player, in turn, signs his or her name on a previously unsigned face. The winner is the player who first succeeds in signing three faces that share a common vertex.

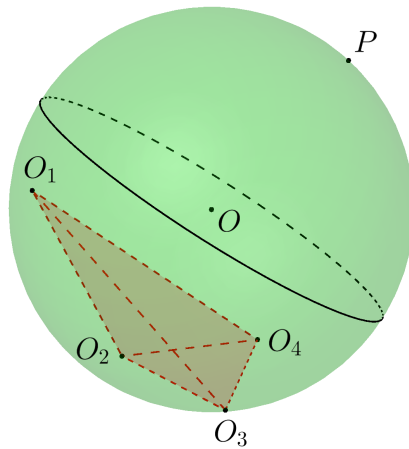
Show that the player who signs first will always win by playing as well as possible.

Solution. We first prove that there exists a face with at least 4 sides. Suppose not. Then each face is a triangle. Let V, E, F be the number of vertices, number of edges and number of faces respectively. Since each vertex is the endpoint of exactly 3 edges, we have $3V = 2E$. Since each face has exactly 3 sides, we have $3F = 2E$. Thus, $V = F$ and $E = \frac{3F}{2}$. By Euler's polyhedron formula, we have $V - E + F = 2$, i.e. $F - \frac{3F}{2} + F = 2$. This yields $F = 4$, which is a contradiction.

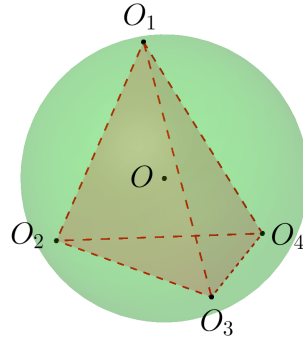
Now, suppose the first player signs on a face F_1 with at least 4 sides. No matter where the second player signs, there must be 3 unsigned faces F_2, F_3, F_4 adjacent to F_1 such that the common edges of these faces with F_1 are consecutive. Since each vertex is the endpoint of exactly 3 edges, F_3 must be adjacent to F_2 and F_4 . The first player can sign on F_3 in the second round. In the third round, the first player must be able to sign on F_2 or F_4 to win the game. Obviously, the second player cannot win before the third round.

Problem 23. A sphere is completely covered by a finite number of (possibly overlapping) hemispheres. Prove that we can choose at most 4 of these hemispheres which together cover the sphere completely.

Solution. Let O be the centre of the sphere. Let $O_1, O_2, \dots, O_n$ be the top points of the hemispheres. Here, the top point means the point on the hemisphere which is furthest away from the base of the hemisphere. For simplicity, we use $H(P)$ to denote the hemisphere with top point P . If two of the O_i 's are antipodes, say O_1 and O_2 , then we can simply take $H(O_1)$ and $H(O_2)$. So we can assume no two of the O_i 's are antipodes.



We claim that O lies inside the convex hull formed by $O_1, O_2, \dots, O_n$. If not, all O_i 's lie on the same side of a great circle. Let P be the point on the sphere which is furthest away from this great circle and lies on different side as the O_i 's. Then $H(P)$ does not contain any of the O_i 's, and hence P is not contained in any $H(O_i)$. This is a contradiction, and the claim is proved.



Since the convex hull of $O_1, O_2, \dots, O_n$ is just a convex polyhedron, and a convex polyhedron can be partitioned into some tetrahedrons, the interior point O must lie inside one of the tetrahedrons $O_1O_2O_3O_4$. Then $H(O_1), H(O_2), H(O_3), H(O_4)$ is a feasible choice. Indeed, if a point P on the sphere is not contained in any of these hemispheres, then O_1, O_2, O_3, O_4 are not contained in $H(P)$. But this is impossible since O lies inside $O_1O_2O_3O_4$. So every point on the sphere is contained in some $H(O_i)$.

Problem 24. There is a set S of $n \geq 4$ points in $\mathbb{R}^3$ such that no 4 points are coplanar. Suppose we can colour all the points in S black and white such that for every sphere passing through at least 4 points in S , exactly half of these common points are coloured white. Prove that all points in S are cospherical.

Solution. Suppose there are b black points and w white points. WLOG assume $b \geq w$. By picking any 4 points, we can find at least two black points and two white points on their circumsphere. Therefore, we can find two black points A, B and a white point C . Consider all spheres passing through A, B, C and at least one more point in S . Suppose there are k such spheres.

On each such sphere, there is one more white point than black point if we ignore the points A, B, C . In total, there are k more white points than black points on these spheres other than A, B, C . As each point except A, B, C lies on exactly one such sphere, we have $w - b = k + 1 - 2 = k - 1$. But then we have assumed $w - b \leq 0$. Therefore, $k \leq 1$. As $n \geq 4$, there exists such a sphere. Thus, $k = 1$. This means all points lie on the same sphere.